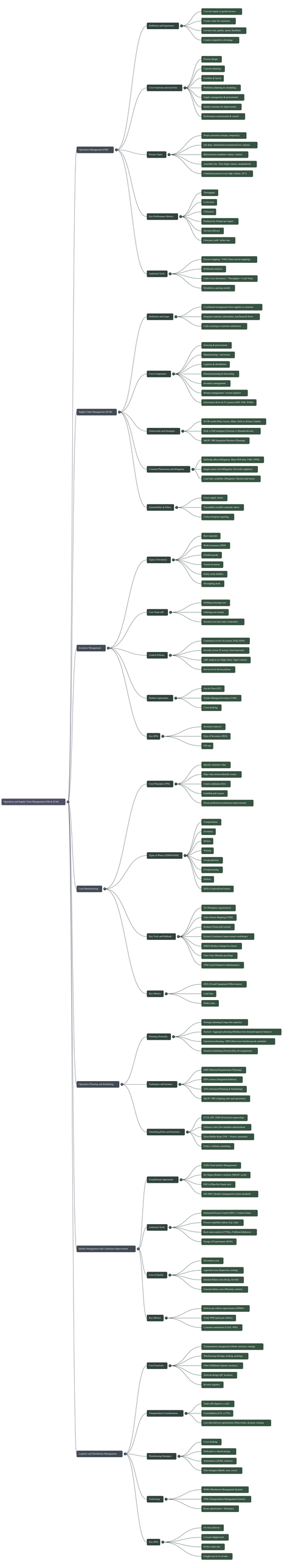

- Operations and Supply Chain Management (OM & SCM)
  - Operations Management (OM)
    - Definition and Importance
      - Converts inputs to goods/services
      - Creates value for customers
      - Governs cost, quality, speed, flexibility
      - Creates competitive advantage
    - Core Functions and Activities
      - Process design
      - Capacity planning
      - Facilities & layout
      - Workforce planning & scheduling
      - Supply management & procurement
      - Quality assurance & improvement
      - Performance measurement & control
    - Process Types
      - Project processes (unique, temporary)
      - Job shop / intermittent (customized low-volume)
      - Batch process (medium volume, variety)
      - Assembly line / flow (high volume, standardized)
      - Continuous process (very high volume, 24/7)
    - Key Performance Metrics
      - Throughput
      - Cycle time
      - Utilization
      - Productivity (Output per input)
      - On-time delivery
      - First-pass yield / defect rate
    - Analytical Tools
      - Process mapping / VSM (Value-stream mapping)
      - Bottleneck analysis
      - Little's Law (Inventory = Throughput × Lead Time)
      - Simulation, queuing models
  - Supply Chain Management (SCM)
    - Definition and Scope
      - Coordinated management from supplier to customer
      - Integrates material, information, and financial flows
      - Links sourcing to customer satisfaction
    - Core Components
      - Sourcing & procurement
      - Manufacturing / conversion
      - Logistics & distribution
      - Demand planning & forecasting
      - Inventory management
      - Returns management / reverse logistics
      - Information flows & IT systems (ERP, TMS, WMS)
    - Frameworks and Strategies
      - SCOR model (Plan, Source, Make, Deliver, Return, Enable)
      - Push vs Pull strategies (Forecast vs Demand-driven)
      - S&OP / IBP (Integrated Business Planning)
    - Common Phenomena and Mitigation
      - Bullwhip effect (Mitigation: Share POS data, VMI, CPFR)
      - Single-source risk (Mitigation: Diversify suppliers)
      - Lead time variability (Mitigation: Shorten lead times)
    - Sustainability & Ethics
      - Green supply chains
      - Traceability (conflict minerals, labor)
      - Carbon footprint reporting
  - Inventory Management
    - Types of Inventory
      - Raw materials
      - Work-in-process (WIP)
      - Finished goods
      - Transit inventory
      - Safety stock (buffer)
      - Decoupling stock
    - Cost Trade-offs
      - Holding (carrying) cost
      - Ordering cost (setup)
      - Stockout cost (lost sales, backorder)
    - Control Policies
      - Continuous review (Q system, EOQ, ROP)
      - Periodic review (P system, fixed intervals)
      - ABC analysis (A=High Value, Tight Control)
      - Service-level driven policies
    - Modern Approaches
      - Just-In-Time (JIT)
      - Vendor-Managed Inventory (VMI)
      - Cross-docking
    - Key KPIs
      - Inventory turnover
      - Days of Inventory (DOI)
      - Fill rate
  - Lean Manufacturing
    - Core Principles (TPS)
      - Specify customer value
      - Map value stream (identify waste)
      - Create continuous flow
      - Establish pull system
      - Pursue perfection (continuous improvement)
    - Types of Waste (TIMWOODS)
      - Transportation
      - Inventory
      - Motion
      - Waiting
      - Overproduction
      - Overprocessing
      - Defects
      - Skills (Underutilized talent)
    - Key Tools and Methods
      - 5S (Workplace organization)
      - Value Stream Mapping (VSM)
      - Kanban (Visual pull system)
      - Kaizen (Continuous improvement workshops)
      - SMED (Reduce changeover times)
      - Poka-Yoke (Mistake-proofing)
      - TPM (Total Productive Maintenance)
    - Key Metrics
      - OEE (Overall Equipment Effectiveness)
      - Lead time
      - Defect rates
  - Operations Planning and Scheduling
    - Planning Hierarchy
      - Strategic planning (Long-term capacity)
      - Tactical / Aggregate planning (Medium-term demand/capacity balance)
      - Operational planning / MPS (Short-term finished goods schedule)
      - Detailed scheduling (Hourly/daily job assignment)
    - Techniques and Systems
      - MRP (Material Requirements Planning)
      - ERP systems (Integrated database)
      - APS (Advanced Planning & Scheduling)
      - S&OP / IBP (Aligning sales and operations)
    - Scheduling Rules and Heuristics
      - FCFS, SPT, EDD (Prioritized sequencing)
      - Johnson's rule (Two-machine optimization)
      - Drum-Buffer-Rope (TOC - Protect constraint)
      - Finite vs Infinite scheduling
  - Quality Management and Continuous Improvement
    - Foundational Approaches
      - TQM (Total Quality Management)
      - Six Sigma (Reduce variation, DMAIC cycle)
      - PDCA (Plan-Do-Check-Act)
      - ISO 9001 (Quality management system standard)
    - Analytical Tools
      - Statistical Process Control (SPC) / Control Charts
      - Process capability indices (Cp, Cpk)
      - Root cause analysis (5 Whys, Fishbone/Ishikawa)
      - Design of Experiments (DOE)
    - Cost of Quality
      - Prevention costs
      - Appraisal costs (Inspection, testing)
      - Internal failure costs (Scrap, rework)
      - External failure costs (Warranty, returns)
    - Key Metrics
      - Defects per million opportunities (DPMO)
      - Yield, PPM (parts per million)
      - Customer satisfaction (CSAT, NPS)
  - Logistics and Distribution Management
    - Core Functions
      - Transportation management (Mode selection, routing)
      - Warehousing (Storage, picking, packing)
      - Order fulfillment (Speed, accuracy)
      - Network design (DC location)
      - Reverse logistics
    - Transportation Considerations
      - Trade-offs (Speed vs. cost)
      - Consolidation (LTL vs FTL)
      - Last-mile delivery optimization (Micro-hubs, dynamic routing)
    - Warehousing Strategies
      - Cross-docking
      - Dedicated vs shared storage
      - Automation (AS/RS, robotics)
      - Pick strategies (Batch, zone, wave)
    - Technology
      - WMS (Warehouse Management System)
      - TMS (Transportation Management System)
      - Route optimization / Telematics
    - Key KPIs
      - On-time delivery
      - Cost per shipped unit
      - Perfect order rate
      - Freight cost as % of sales